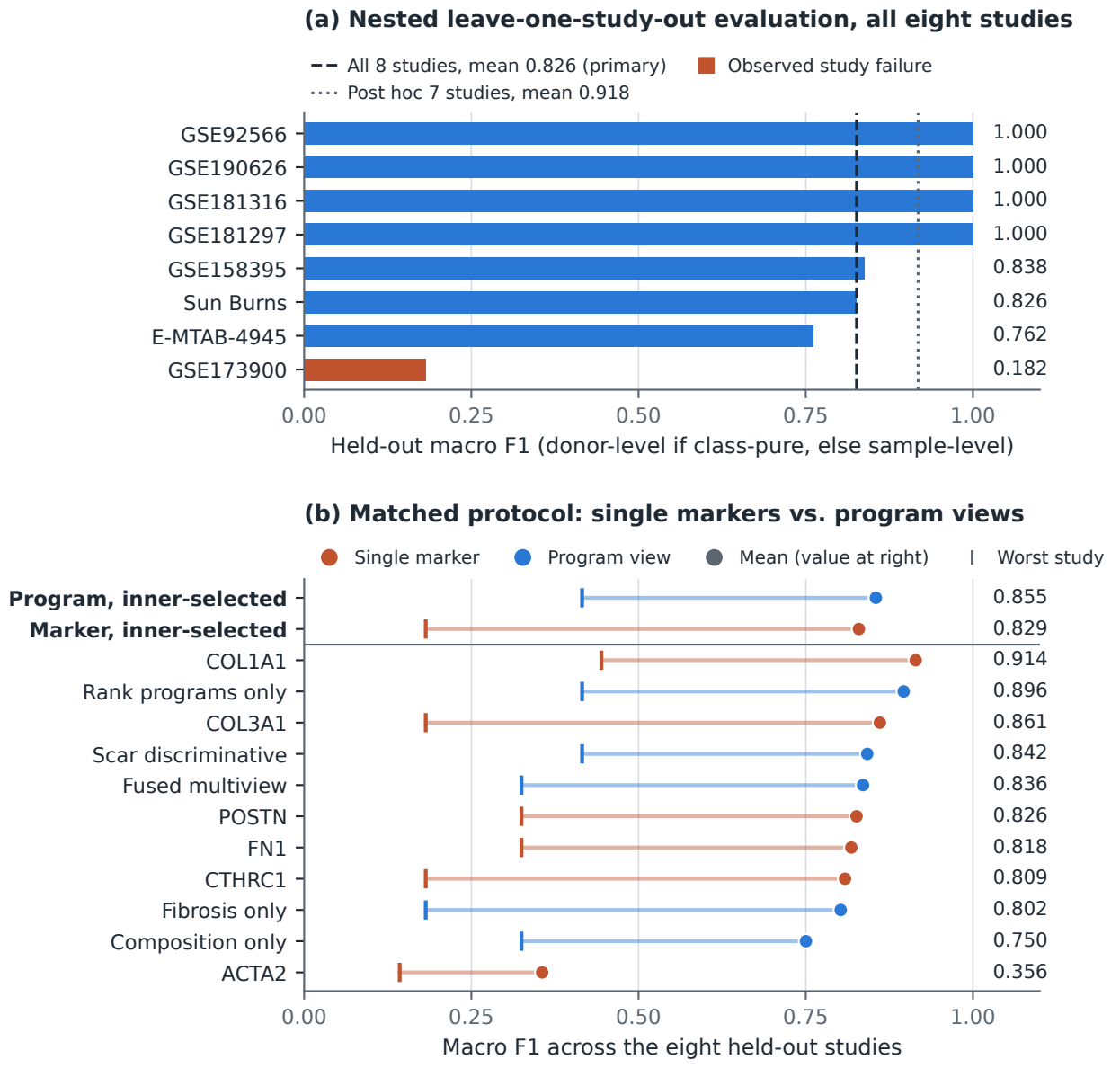
(a) Nested leave-one-study-out evaluation, all eight studies
All 8 studies, mean 0.826 (primary)
Observed study failure
Post hoc 7 studies, mean 0.918
GSE92566
GSE190626
GSE181316
GSE181297
GSE158395
Sun Burns
E-MTAB-4945
GSE173900
1.000
1.000
1.000
1.000
0.838
0.826
0.762
0.182
0.00
0.25
0.50
0.75
1.00
Held-out macro F1 (donor-level if class-pure, else sample-level)
(b) Matched protocol: single markers vs. program views
Single marker
Program view
Mean (value at right)
Worst study
Program, inner-selected
Marker, inner-selected
COL1A1
Rank programs only
COL3A1
Scar discriminative
Fused multiview
POSTN
FN1
CTHRC1
Fibrosis only
Composition only
ACTA2
0.855
0.829
0.914
0.896
0.861
0.842
0.836
0.826
0.818
0.809
0.802
0.750
0.356
0.00
0.25
0.50
0.75
1.00
Macro F1 across the eight held-out studies